EE2001 Electrical Circuits- I:

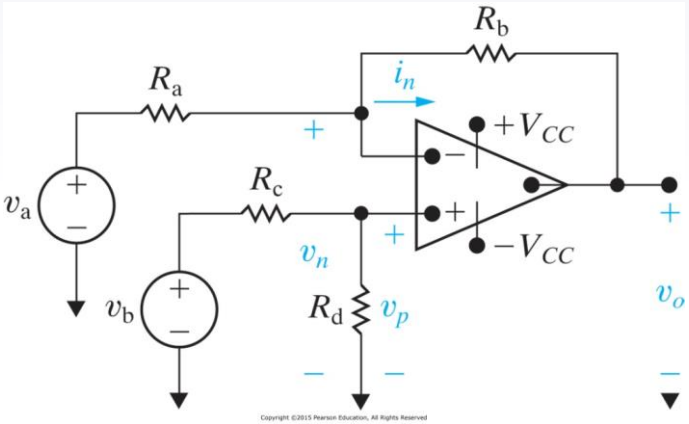
Continuation of the topic:
The Operational Amplifier

12 December 2023

The Difference Amplifier Circuit

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- Example:
- a) Design a difference amplifier that amplifies the difference between two input voltages by a gain of 8, using an ideal op-amp and $\pm 8\ V$ power supplies.



- Using the simplified difference-amplifier equation:

$$v_o = \frac{R_b}{R_a}(v_b - v_a) = 8(v_b - v_a) \quad \text{so} \quad \frac{R_b}{R_a} = 8$$

We want two resistors whose ratio is 8. We can choose $R_b = 80\ k\Omega$ and $R_a = 10\ k\Omega$ although there are many other possibilities.

Note that simplified difference-amplifier equation requires that,

$$\frac{R_a}{R_b} = \frac{R_c}{R_d}$$

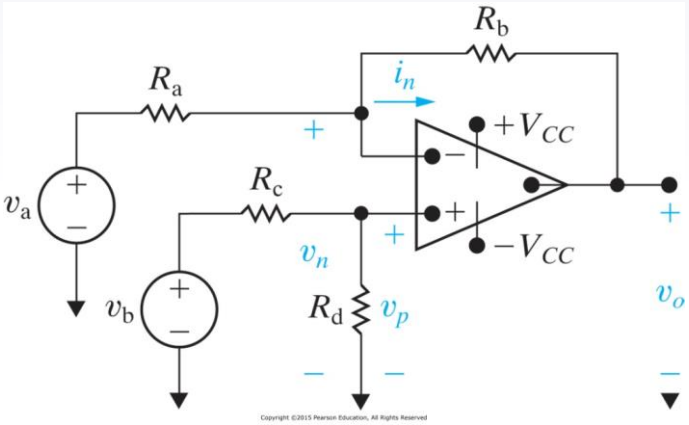
A simple choice would be $R_c = 15\ k\Omega$ and $R_d = 120\ k\Omega$

The Difference Amplifier Circuit

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- Example:

b) Suppose $v_a = 1\,V$ in the difference amplifier designed in part (a). What range of input voltages for v_b can allow the op-amp to remain in its linear region?



- Solve two different version of the simplified difference-ampifier equation for v_o in terms of v_b, first using $v_0 = +8V$ and then using $v_0 = -8V$

$v_o = 8(v_b - 1) = 8\,V$ so $v_b = 2V$

$v_o = 8(v_b - 1) = -8\,V$ so $v_b = 0V$

Thus, if $v_a = 1\,V$ in the difference-amplifier from part (a), the op-amp will remain in its linear region of operation if $0 \leq v_b \leq +2V$

The Common and Difference Modes in the Difference Amplifier

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- The differential mode of an amplifier is defined as,

$$v_{dm} = v_b - v_a$$

- The common mode of an amplifier is defined as,

$$v_{cm} = \frac{v_a + v_b}{2}$$

- In terms of these variables, one can express v_a and v_b as,

$$v_a = v_{cm} - \frac{1}{2}v_{dm} \qquad v_b = v_{cm} + \frac{1}{2}v_{dm}$$

- Using these the output voltage of the difference amplifier can be expressed as (after some derivations),

$$v_o = \underbrace{\left(\frac{R_aR_d - R_bR_c}{R_a(R_c + R_d)}\right)}_{A_{cm}} v_{cm} + \underbrace{\left(\frac{R_d(R_a + R_b) + R_b(R_c+R_d)}{2R_a(R_c + R_d)}\right)}_{A_{dm}} v_{dm}$$

The Common and Difference Modes in the Difference Amplifier

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$$v_o = A_{cm}v_{cm} + A_{dm}v_{dm}$$

- A_{cm}= the common mode gain
- A_{dm}= the differential mode gain

$$A_{cm} = \frac{R_aR_d - R_bR_c}{R_a(R_c+R_d)} \qquad A_{dm} = \frac{R_d(R_a+R_b)+R_b(R_c+R_d)}{2R_a(R_c+R_d)}$$

- For $\frac{R_a}{R_b} = \frac{R_c}{R_d}$

  we have

$$A_{cm} = 0 \qquad A_{dm} = \frac{R_b}{R_a}$$

- Then the output voltage is given by

$$v_o = 0.\, v_{cm} + \frac{R_b}{R_a} v_{dm}$$

The Common and Difference Modes in the Difference Amplifier

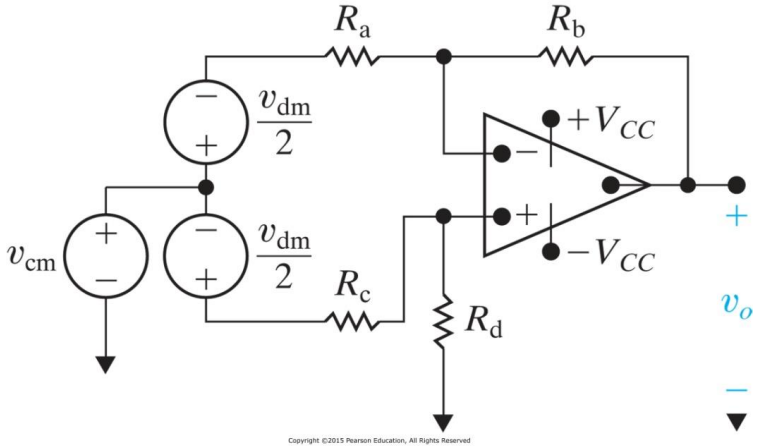
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- The ideal difference amplifier has $A_{cm} = 0$.

  - It amplifies only the differential mode portion of the input voltage and eliminates the common mode portion of the input voltage.

- In many applications the differential mode portion contains information of interest, whereas the common mode portion contains noise and needs to be eliminated.

- In terms of v_{cm} and v_{dm} the circuit can be redrawn as



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The Common Mode Rejection Ratio

- In practice, the common mode gain A_{cm} of the difference amplifier may not be zero due to
  - Resistance mismatch
  - Non-ideal op-amp
- Lets consider the effect of resistance mismatches.
- Assume that the chosen resistors do not satisfy the below condition exactly.

$$\frac{R_a}{R_b} = \frac{R_c}{R_d}$$

- Instead, assume that the following relation is satisfied (due to resistance mismatch), where ε is a very small constant.

$$\frac{R_a}{R_b} = (1 - \varepsilon)\frac{R_c}{R_d}$$

- Under this condition, the new expressions for the common mode gain A_{cm} and the differential mode gain A_{dm} can be derived as follows.

The Common Mode Rejection Ratio

- Under this condition the new expressions for the common mode gain A_{cm} and the differential mode gain A_{dm} can be derived as follows.

$$A_{cm} \cong \frac{-\varepsilon R_b}{R_a+R_b} \qquad A_{dm} \cong \frac{R_b}{R_a}\left(1-\frac{(\varepsilon/2)R_a}{R_a+R_b}\right)$$

- Note that if $\varepsilon = 0$, then we have $A_{cm} = 0$ and $A_{dm} = \frac{R_b}{R_a}$ (as we had before)
- The Common Mode Rejection Ratio (CMMR) is defined as:

$$CMMR = \left|\frac{A_{dm}}{A_{cm}}\right|$$

- CMMR can be used as a measure how nearly ideal the amplifier is.
  - The higher the CMMR, the closer to ideal the amplifier is.

The Common Mode Rejection Ratio

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- By substituting the obtained values for the common mode gain A_{cm} and the differential mode gain A_{dm} the CMMR for the difference amplifier under resistnce mismatch becomes,

$$CMMR = \left|\frac{A_{dm}}{A_{cm}}\right| \cong \left|\frac{\frac{R_b}{R_a}\left(1 - \frac{\left(\frac{\varepsilon}{2}\right)R_a}{R_a + R_b}\right)}{\frac{-\varepsilon R_b}{R_a + R_b}}\right| \cong \cdots \cong \left|\frac{1 + \frac{R_b}{R_a}}{-\varepsilon}\right|$$

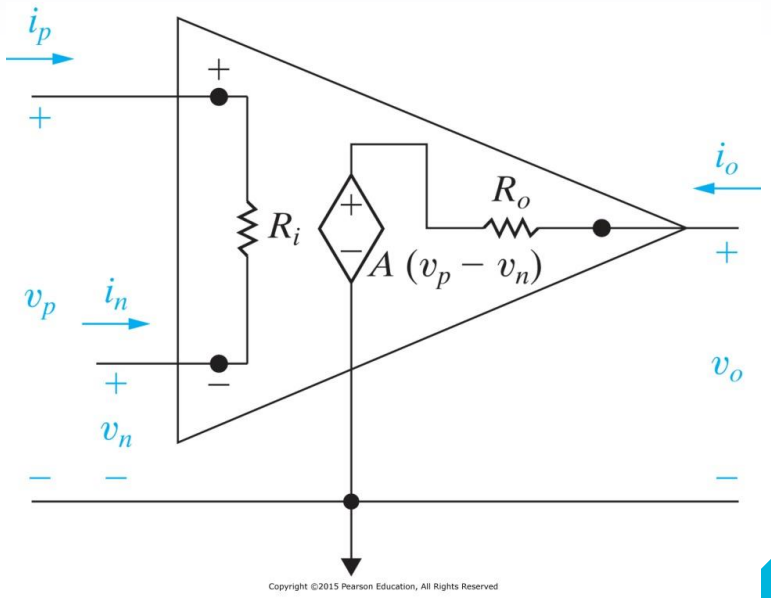
- Note that when $\varepsilon = 0,\ CMMR \to \infty$
- Even if the resistors mismatch, by increasing the difference mode gain R_b/R_a the CMRR can be improved. Note also that the op-amp itself is a difference amplifier.
- Therefore, it also has common mode and differential mode outputs (which are caused by internal component mismatch in the op-amp). The CMRR of the op-amp can be used to assess how ideal the op-amp is. The non-ideal op-amp can affect the output of the above discussed different amplifier as well.

A More Realistic Model for the Operational Amplifier

- The ideal op-amp model is a simplified model for the operation of the operational amplifier.
- There is also a more realistic model which we will discuss below.
- This model includes three modifications to the ideal op-amp model:
- Finite input resistance R_i
- Finite open loop gain A
- Non-zero output resistance R_o
- This model can be represented as in the figure.
- The assumptions for the ideal op-amp do not hold anymore.

$$i_p = i_n = 0 \qquad v_p = v_n \qquad v_o = A(v_p - v_n)$$

Do not hold anymore.

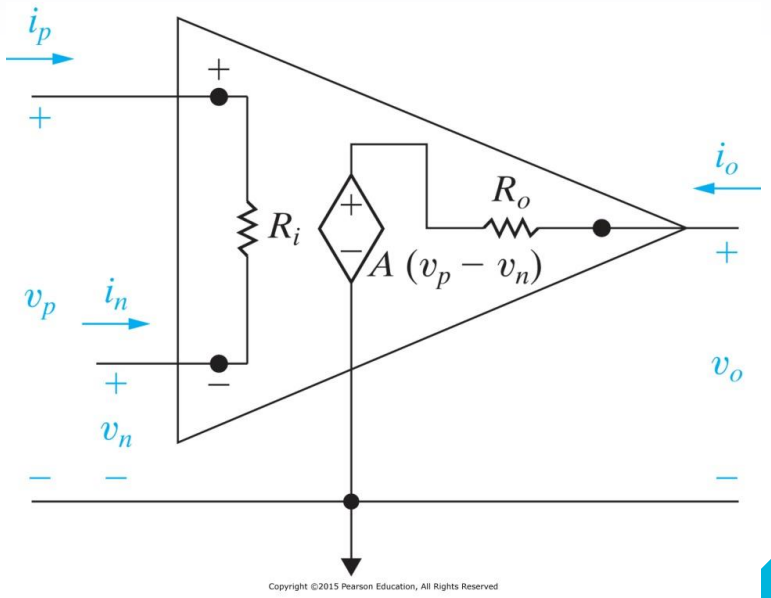


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A More Realistic Model for the Operational Amplifier

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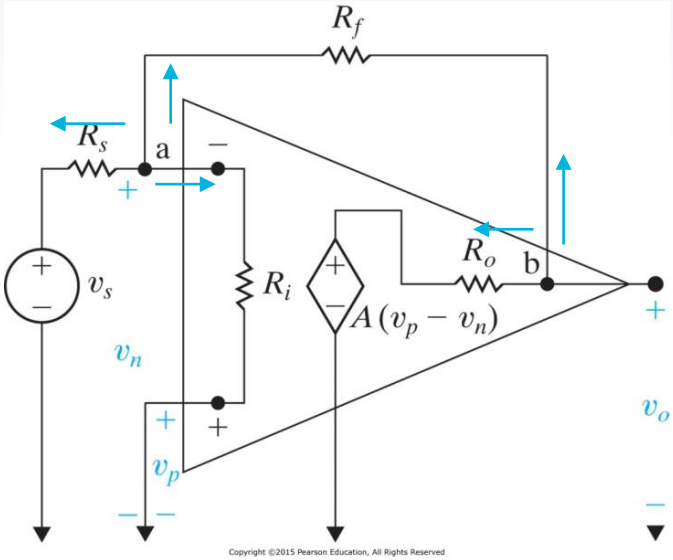
- Note that as $R_i \to \infty$, $A \to \infty$, and $R_o \to 0$ the circuit for the more realistic model reduces to the ideal op-amp.
- For the μA741 op-amp, typical values for these are $R_i = 2M\Omega$, A= 10^5, and $R_o = 75\Omega$.
- Analysis of op-amp circuits using the more realistic model is still straightforward (with little bit more calculations).



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Inverting Amplifier Circuit with the More Realistic Op-Amp Model

- Consider the inverting amplifier



$$v_p = 0$$

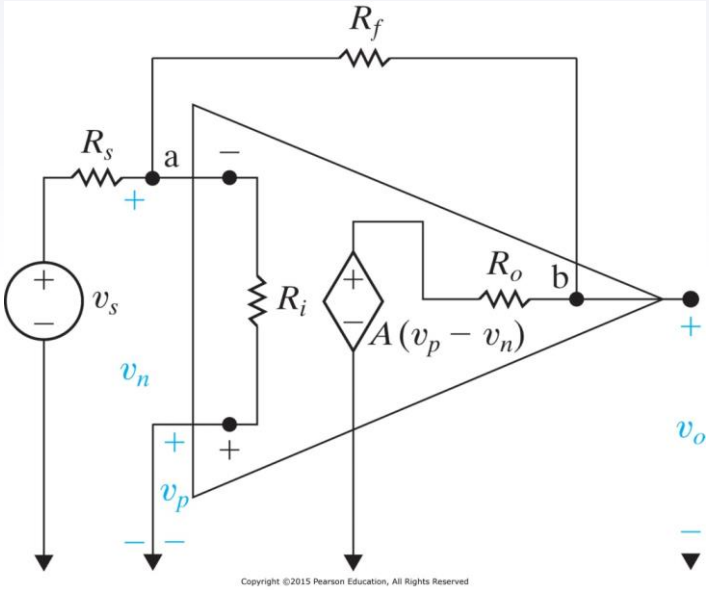
$$\frac{v_n - v_s}{R_s} + \frac{v_n - v_p}{R_i} + \frac{v_n - v_o}{R_f} = 0$$

$$\frac{v_o - v_n}{R_f} + \frac{v_o - A(v_p - v_n)}{R_o} = 0$$

- Solving these equations for v_o, one obtains,

$$v_o = \frac{-A + \frac{R_o}{R_f}}{\frac{R_s}{R_f}\left(1 + A + \frac{R_o}{R_i}\right) + \left(\frac{R_s}{R_i} + 1\right) + \frac{R_o}{R_f}} v_s$$

Inverting Amplifier Circuit with the More Realistic Op-Amp Model



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- Note that as $R_o \to 0$, $R_i \to \infty$, and $A \to \infty$ we have,

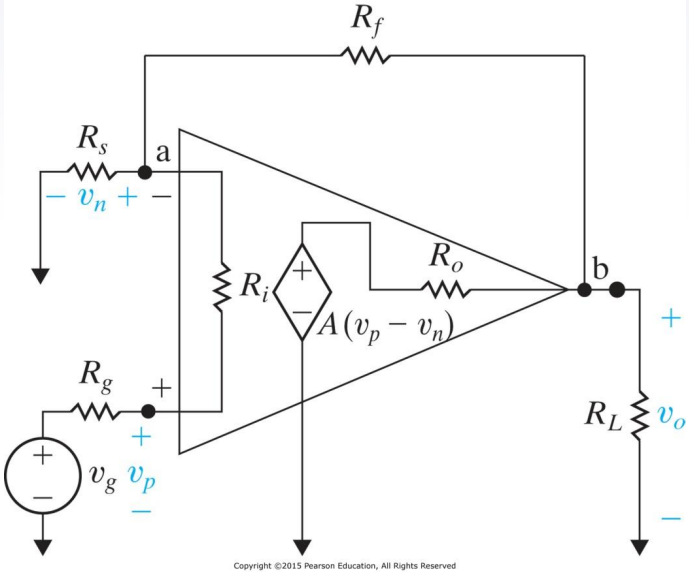
$$v_o \to \frac{-A}{\frac{R_s}{R_f}A} v_s = -\frac{R_f}{R_s} v_s$$

- In case there is a load resistor R_L connected to the output of the op-amp, it needs to be considered as well and the output voltage equation becomes.

$$v_o = \frac{-A + \frac{R_o}{R_f}}{\frac{R_s}{R_f}\left(1 + A + \frac{R_o}{R_i} + \frac{R_o}{R_L}\right) + \left(1 + \frac{R_o}{R_L}\right)\left(\frac{R_s}{R_i} + 1\right) + \frac{R_o}{R_f}} v_s$$

Non-Inverting Amplifier Circuit with the More Realistic Op-Amp Model

- Consider the non-inverting amplifier



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$$\frac{v_n}{R_s} + \frac{v_n - v_p}{R_i} + \frac{v_n - v_o}{R_f} = 0$$

$$\frac{v_o - v_n}{R_f} + \frac{v_o - A(v_p - v_n)}{R_o} + \frac{v_o}{R_L} = 0$$

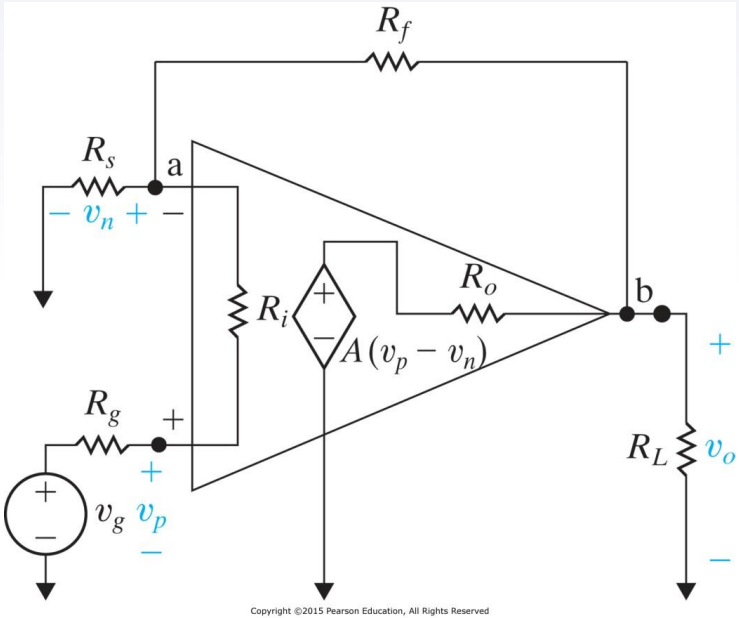
$$\frac{v_p - v_g}{R_g} = \frac{v_n - v_p}{R_i}$$

- Solving these equations for v_o, one obtains,

$$v_o = \frac{\left[(R_f + R_s) + \left(\frac{R_s R_o}{AR_i}\right)\right]}{R_s + \frac{R_o}{A}(1 + K_r) + \frac{R_f R_s + (R_f + R_s)(R_i + R_g)}{AR_i}} v_g$$

$$K_r = \frac{R_s + R_g}{R_i} + \frac{R_f + R_s}{R_L} + \frac{R_f R_s + R_f R_g + R_g R_s}{R_i R_L}$$

Non-Inverting Amplifier Circuit with the More Realistic Op-Amp Model



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- Note that as $R_o \to 0$, $R_i \to \infty$, and $A \to \infty$ we have,

$$v_o \to \left(1 + \frac{R_f}{R_s}\right) v_g$$

- In case there is a load resistor R_L is omitted from the output, the output voltage equation becomes.

$$v_o = \frac{\left[(R_f + R_s) + \left(\frac{R_s R_o}{A R_i}\right)\right]}{R_s + \frac{R_o}{A}\left(1 + \frac{R_s + R_g}{R_i}\right) + \frac{R_f R_s + (R_f + R_s)(R_i + R_g)}{A R_i}} v_g$$

End of the Lesson (12.12.23)

Any questions?

New supplementary videos are added to the playlist on Youtube:

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist